SPEEDING ON THE DIGITAL FREEWAY

Be it quantum computing or AI, Indian govt and private players are taking up the development of new-age tech on mission mode. There can't be any let-up now

By **Ajay Sukumaran**

At **the Tata Institute of Fundamental Research (TIFR) in Mumbai,** work to build India's first quantum computer is nearing completion. When ready, this small-scale quantum computer will be a significant milestone for TIFR's Quantum Measurement and Control (QuMaC) lab, established 12 years ago to address fundamental challenges in building quantum systems. Dr R. Vijayaraghavan, who heads QuMaC, sees the project as a crucial first step for India that will "allow us to get into this game". The project they are working on, in collaboration with the Defence Research and Development Organisation (DRDO) and Tata Consultancy Services (TCS), involves designing key components, such as the quantum processing unit, electronics and software—each of which presents multiple levels of complexity.

A quantum bit or qubit is the basic unit of a quantum computing system. Think of it as the equivalent of bits, building blocks of the conventional computers. But while a bit exists in binaries (of either 0 or 1), a qubit can hold both values at the same time. This ability to be in multiple states simultaneously is known as superposition, which means a quantum computer can theoretically tackle far more complex calculations than even the most powerful supercomputer today. For example, it could easily break all the secure algorithmic codes that we rely on today for banking transactions, military communication etc. Or take the case of drug discovery, where there's a need to simulate molecules in various states, or supply-chain logistics, where the goal is to arrive at the most optimal procedure—quantum computers could offer a disruptive advantage in carrying out all such functions. That's the reason quantum computing, along with Artificial Intelligence (AI), features on the list of 'critical' technologies on the national agendas of most countries. Globally, funding announcements into quantum technologies by various governments over the years touched $42 billion (Rs 3.5 lakh crore, at the current exchange rate) in 2023, according to McKinsey's Quantum Technology Monitor published in April this year. India, too, announced a National Quantum Mission in April 2023, for which Rs 6,000 crore has been earmarked up to 2030-31. "Quantum technology is the new technology frontier, reached after decades of fundamental research," said Professor Ajay Kumar Sood, principal scientific advisor to the Government of India, in a release issued on April 14, celebrated as World Quantum Day.

THE FIRST STEP

For QuMaC and their collaborators, the small-scale quantum computer is a technology demonstration. Whatever equations it can solve can be "perfectly mimicked" by a laptop, says Vijayaraghavan. "But if we have to build a 100-qubit system, we have to start somewhere. So, this gives us all the necessary expertise in understanding the different parts of a quantum computer and where the challenges are." He points to how IBM, about eight years ago, launched its first 5-qubit quantum computer on the cloud. "For us," he adds, "it's the first step." As part of the National Quantum Mission, Vijayaraghavan leads a team of eight scientists from across five institutes which is putting together a proposal to build a 24-qubit computer in three years and a 100-qubit computer in five years. Likewise, there are other teams of specialists working on different proposals—these include various technological approaches to building quantum systems using photons, neutral atoms or trapped ions as qubits.

However, scientific challenges abound because qubits are also unstable and vulnerable to the slightest disturbance. Globally, the challenge before researchers is to find ways around this 'decoherence' via error-corrected qubits. "You have to show that by using such a system, you are actually solving some problem which is of relevance to industry or science or society and show that it is better, faster and cheaper," says Vijayaraghavan. "That of course will be the first holy grail of useful quantum computers. We are not there yet."

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